

The Global Trends of Energy Efficiency and Its Role in Climate Change Mitigation

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1. Introduction

Energy efficiency refers to how much economic output a country can produce from a given amount of energy. It has become increasingly important in recent years as governments try to balance economic growth with efforts to reduce carbon emissions and energy costs. Understanding how energy efficiency has changed over time and what drives these changes can help policymakers design more effective strategies.

This report analyses global energy efficiency trends between 1980 and 2019, using a sample of 28 countries worldwide. Energy efficiency is measured using the Energy Efficiency Index (EEI), calculated as Total Energy Consumption divided by Gross Domestic Product (GDP), multiplied by one million to make the numbers easier to work with. A lower EEI means a country is producing more output for each unit of energy consumed, so lower values represent better performance.

The report is structured as follows: Section 2 looks at global EEI trends over time. Section 3 compares developed and developing economies. Section 4 places the United Kingdom within the global picture. Section 5 investigates the economic factors that drive energy efficiency. Section 6 examines the link between energy efficiency and CO_2 emissions. Section 7 analyses Energy Performance Certificate (EPC) scores across England and Wales. Section 8 concludes with a summary of the findings.

2. Global Energy Efficiency Trends (Q1)

Table 1 shows the summary statistics for the global average EEI broken down by decade. The most striking finding is the consistent downward trend: the mean EEI fell from 7.87 in the 1980s to 4.45 in the 2010s, a reduction of around 44%. This tells us that, on average, the countries in our sample have become significantly more energy efficient over the observed period.

Table 1: EEI Summary Statistics by Decade

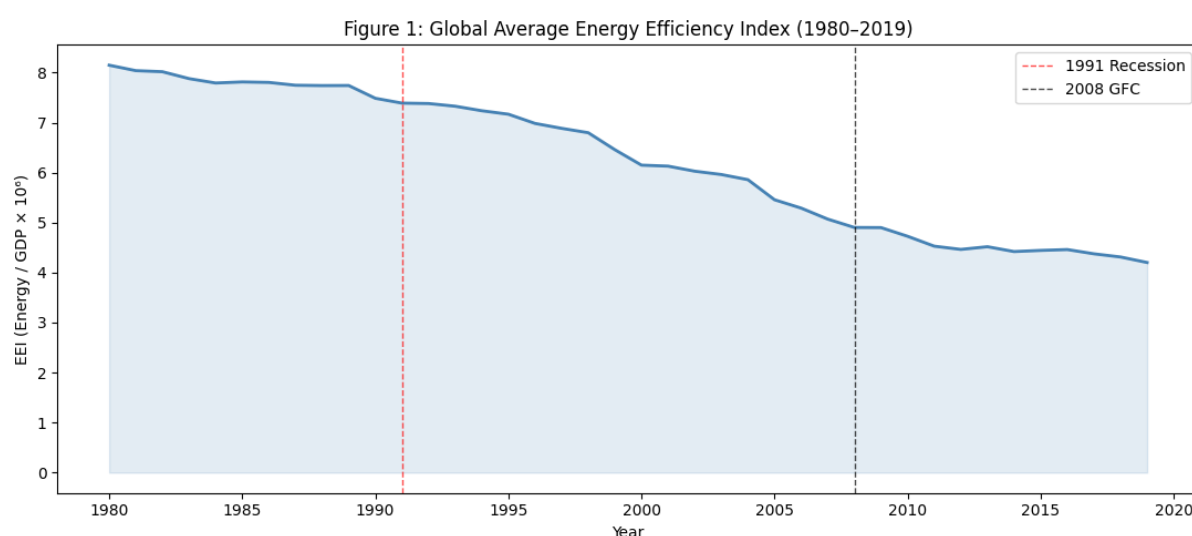
Decade	Mean	Median	Std dev	Min	Max
1980s	7.87	7.15	4.22	1.18	19.34
1990s	7.11	6.54	3.15	1.94	16.53
2000s	5.57	5.21	1.95	0.97	11.45
2010s	4.45	4.01	1.82	1.04	8.79

Note: $EEI = \text{Energy Consumption} / \text{GDP} \times 10^6$. Lower values indicate greater energy efficiency.

Looking more closely at the data, a few things stand out. The mean is consistently higher than the median in every decade, which means a small number of very inefficient countries are pulling the average up while most countries cluster at lower values. The standard deviation also falls over time (from 4.22 to 1.82), suggesting

that countries are becoming more similar in their energy efficiency levels – less efficient countries appear to be gradually catching up with more efficient ones. The maximum value drops sharply from 19.34 to 8.79, again confirming that even the least efficient countries in the sample have made real progress.

Figure 1 plots the global average EEI over time. The decline is fairly smooth and continuous. Two events temporarily interrupted this trend: the 1991 recession and the 2008 Global Financial Crisis (GFC). During both periods, economic output fell faster than energy consumption, causing the EEI to rise briefly before resuming its downward path. This highlights that EEI is sensitive to economic downturns and should be interpreted over the long run rather than year by year.



3. Developed vs Developing Economies (Q2)

3.1 Summary Statistics and Visualisation

Table 2 compares the EEI between developed and developing economies across the full observed period. Developed economies have a higher average EEI (7.29) compared to developing economies (5.11). This might seem surprising at first – you might expect richer countries to be more efficient. However, this reflects the composition of the sample: the developed group includes energy-intensive economies such as Canada and Norway alongside more efficient ones like France and the UK, which drives the average up.

Table 2: EEI Summary Statistics by Development Status (1980-2019)

Group	Mean	Median	Std Dev	Min	Max
Developed	7.29	6.51	3.14	2.54	19.34
Developing	5.11	4.38	2.89	0.97	16.53

Note: Developed = World Bank high-income countries. Developing = all other countries in the sample.

Figure 2a shows how the average EEI for each group has changed over time. Both groups show a clear downward trend, meaning energy efficiency has improved in developed and developing economies alike. Interestingly, developed economies started with a much higher EEI in 1980 (around 11) compared to developing countries (around 6), but by 2019, both groups had converged to similar levels of around 4-5. This convergence suggests that developing economies have been catching up, possibly through adopting technologies and practices from more advanced economies.

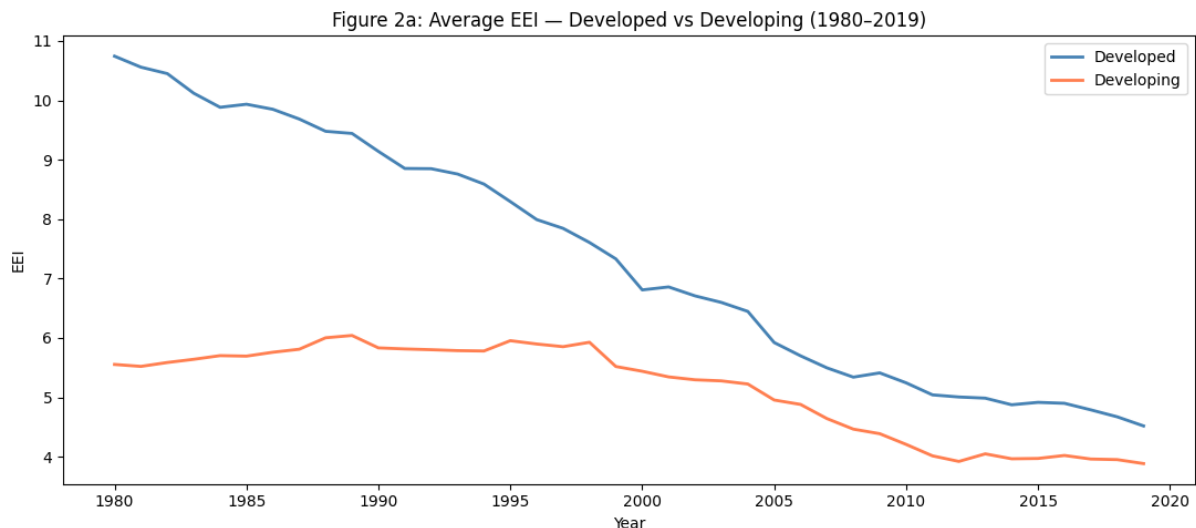
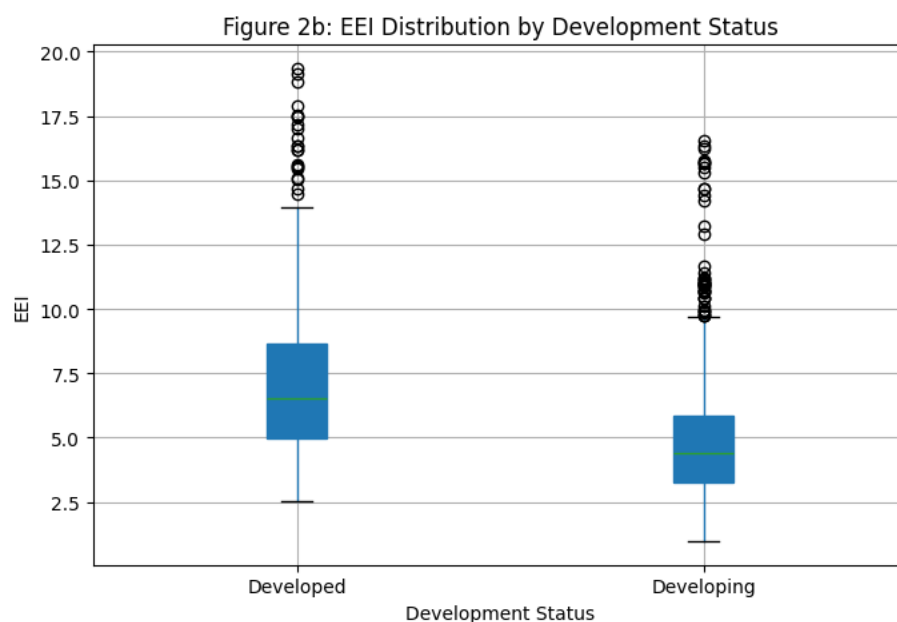


Figure 2b shows box plots of the EEI distribution for both groups. The developed country distribution is more spread out, with a long upper tail driven by countries like Canada and Norway, which have energy-intensive industries and cold climates which require a lot of heating. The developing country distribution is tighter, reflecting a more similar group of economies.



3.2 Regression Analysis

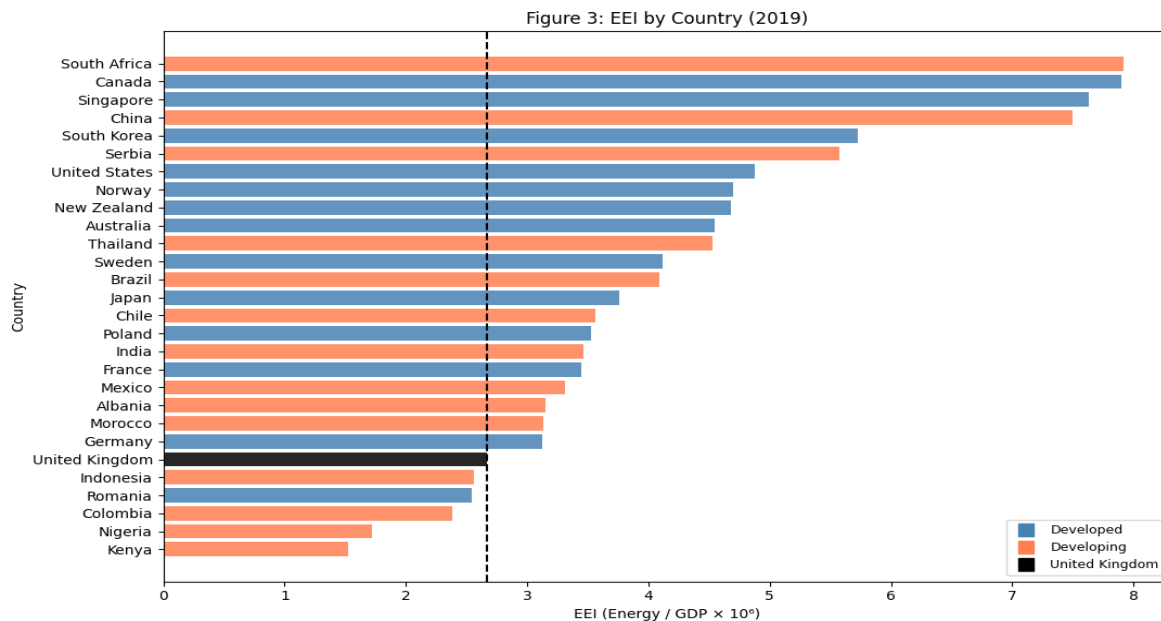
To formally test whether there is a statistically significant difference between the two groups, an OLS regression was run with log EEI as the dependent variable and a dummy variable equal to one for developed economies as the independent variable. The regression uses heteroscedasticity-robust standard errors to ensure the results are reliable.

The coefficient on the development dummy is 0.410 and is highly significant ($p < 0.001$). This means developed economies have a statistically significantly higher log EEI than developing economies, confirming that the difference observed in the descriptive statistics is not due to chance. However, the R^2 of 0.163 indicates that development status alone explains only about 16% of the variation in EEI – other factors, such as country characteristics and time trends, might help explain the variation better.

4. The UK in Global Context (Q3)

In 2019, the United Kingdom's EEI was 2.67, placing it 6th out of 28 countries when ranked from most to least efficient. This means the UK performs better than roughly 79% of the countries in the sample, putting it firmly among the most energy-efficient economies.

Figure 3 presents all 28 countries ranked by their EEI in 2019. Developed economies are shown in blue, developing economies in orange, and the UK is highlighted in black. South Africa and Canada have the highest EEI values (above 7.5), reflecting their reliance on energy-intensive industries such as mining and natural resource extraction. At the other end, Kenya, Nigeria, Colombia, and Indonesia have the lowest EEI values, largely because their economies are less industrialised and consume less energy per unit of output.



The UK's strong performance can be explained by several factors: a large services sector, significant reductions in coal use in favour of renewables and gas, and energy efficiency improvements in buildings and industry driven by government policy. Other large European economies such as Germany and France also appear in the more efficient half of the chart, consistent with their strong energy efficiency policies.

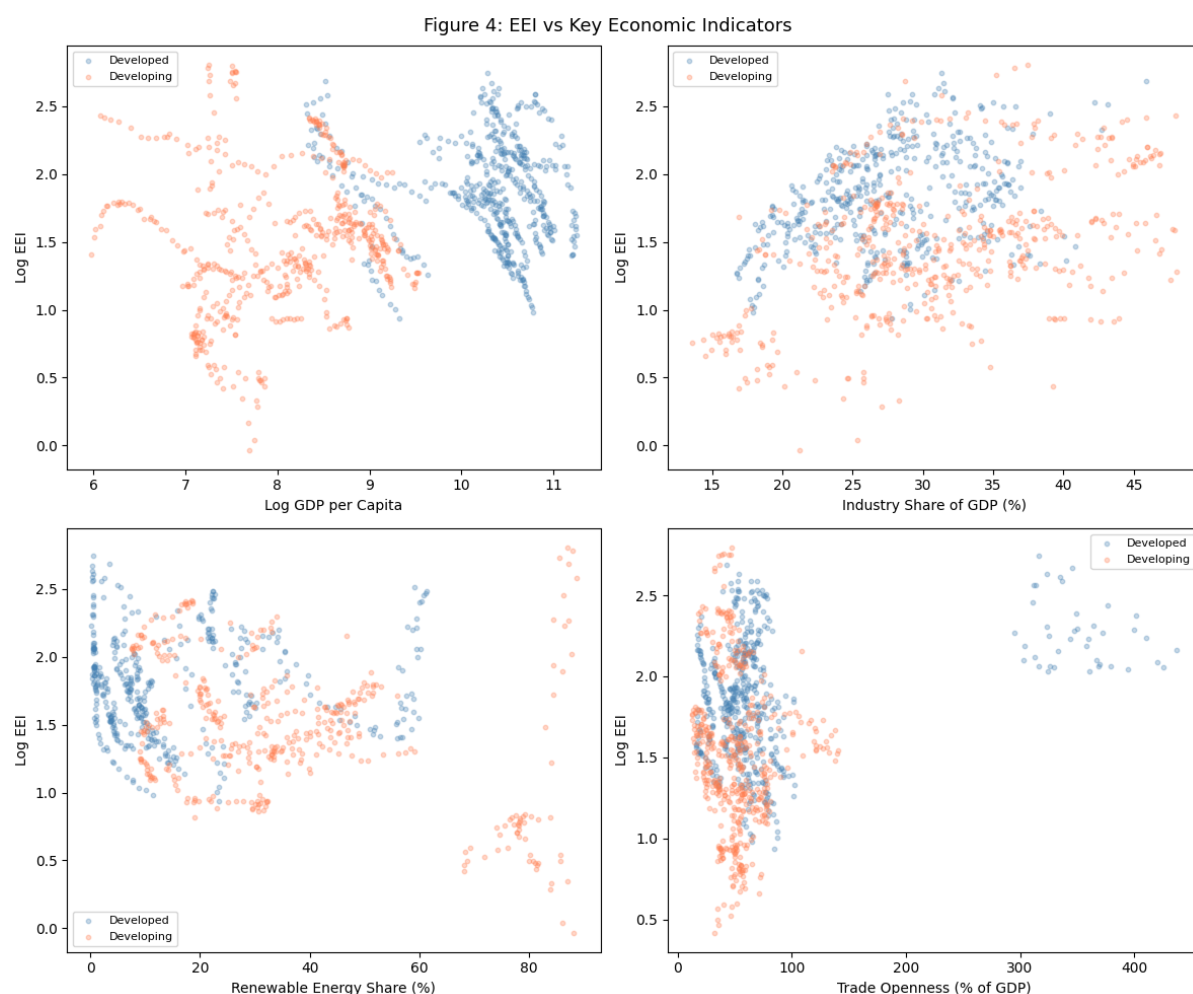
5. Economic Drivers of Energy Efficiency (Q4)

This section looks at the key economic factors that explain differences in EEI across countries and over time. Four indicators from the World Bank are used: log GDP per capita (measuring the level of economic development), industry share of GDP (capturing how much of the economy is manufacturing and heavy industry), renewable energy share (measuring the clean energy mix), and trade openness as a percentage of GDP (measuring how integrated a country is in global trade). A fixed effects regression is used, which includes both country and year dummy variables to control for differences between countries and changes over time that are not captured by the four indicators.

5.1 Scatter Plots

Figure 4 shows scatter plots of log EEI against each of the four indicators, with developed economies in blue and developing in orange. The GDP per capita scatter shows a clear negative relationship – richer countries tend to have lower EEI, meaning they are more energy efficient. The renewables share scatter also shows a negative relationship, particularly among the developed economies: countries with more renewable energy tend to be more efficient. The trade openness scatter is more mixed, and the industry share scatter shows a broadly positive relationship,

suggesting that more industrialised economies tend to use more energy per unit of output.



5.2 Regression Results

Table 3 shows the fixed effects OLS regression results. The high R^2 values (above 0.90 in all three models) indicate that the models explain a very large share of the variation in EEI once country and time differences are accounted for.

Table 3: Fixed Effects OLS Regression – Dependent Variable: Log EEI

Variable	Full Sample	Developed	Developing	Notes
Log GDP per Capita	-0.184***	-0.169	-0.016	Sig. full only
Industry Share GDP	-0.001	-0.013**	0.000	Sig. developed only
Renewable Energy Share	-0.015***	-0.022***	0.002	Sig. full & developed
Trade Openness	0.001**	0.001	0.003***	Sig. full & developing
R^2	0.904	0.890	0.937	

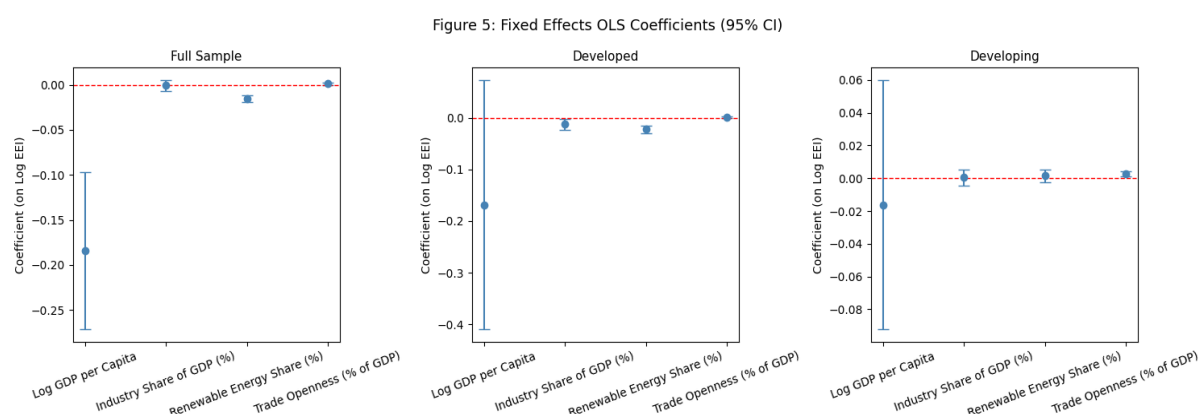
N	765	396	369	
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Note: *** $p < 0.01$, ** $p < 0.05$. Heteroscedasticity-robust standard errors. Country and year dummies included but not reported.

In the full sample, GDP per capita has a negative and significant coefficient of -0.184. This means that as countries become wealthier, they tend to become more energy efficient. Renewable energy share is also negative and significant (-0.015), confirming that countries with a higher share of renewables in their energy mix are more efficient. Trade openness is positive and marginally significant (0.001), suggesting that more trade-open economies are slightly less efficient, possibly due to the energy costs of export-oriented manufacturing.

When the sample is split by development status, interesting differences appear. Among developed economies, renewable energy share remains the strongest driver (-0.022), and industry share of GDP also becomes significant (-0.013) – meaning that as developed economies move away from heavy industry, their energy efficiency improves. GDP per capita is no longer significant within the developed group, which makes sense as all countries in this group are already wealthy.

Among developing economies, only trade openness is significant (0.003). This suggests that for developing countries, being more integrated in global trade is the main factor associated with energy efficiency changes, while income levels and energy mix do not yet play a clear role. The error bar plots in Figure 5 show these patterns visually – the bars for renewable energy share only clearly cross zero (become significant) in the developed subsample.



5.3 Policy Recommendation (Q5)

Based on the regression findings, different policy approaches are needed for developed and developing economies.

For developed economies, the results suggest that increasing the share of renewable energy and continuing to shift away from heavy industry are the most effective ways to improve energy efficiency. Policies such as subsidies for clean energy, energy efficiency standards for buildings and appliances, and carbon pricing mechanisms would all support this. The UK's own policies – including its Emissions

Trading Scheme and investment in offshore wind – provide a useful example for other developed countries to follow (Department for Energy Security and Net Zero, 2023).

For developing economies, the significance of trade openness suggests that access to international markets – and the cleaner technologies that come with them – is an important channel for improving energy efficiency. Policies that make it easier for developing countries to import energy-efficient technology, such as reduced trade tariffs on clean energy equipment, could therefore be effective. International cooperation and technology transfer programmes, such as those supported by the World Bank, could play a key role here (World Bank, 2023).

6. Energy Efficiency and CO_2 emissions

6.1 Descriptive Analysis

Figure 6a shows global average EEI and total CO_2 emissions on the same chart from 1980 to 2019. While the EEI declines steadily over the period – indicating improving efficiency – total CO_2 emissions continue to rise. This might seem contradictory, but it can be explained by the fact that economic growth – particularly in rapidly growing economies like China and India – has led to much higher total energy use, which has outweighed the efficiency gains made. In other words, countries are doing more with each unit of energy, but they are using so many more units overall that emissions have still gone up.

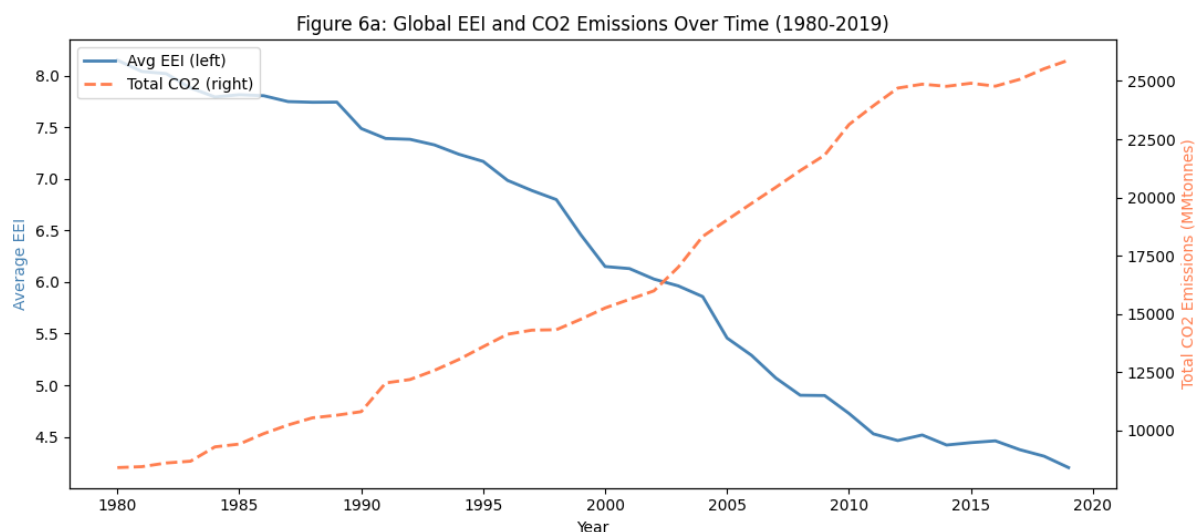
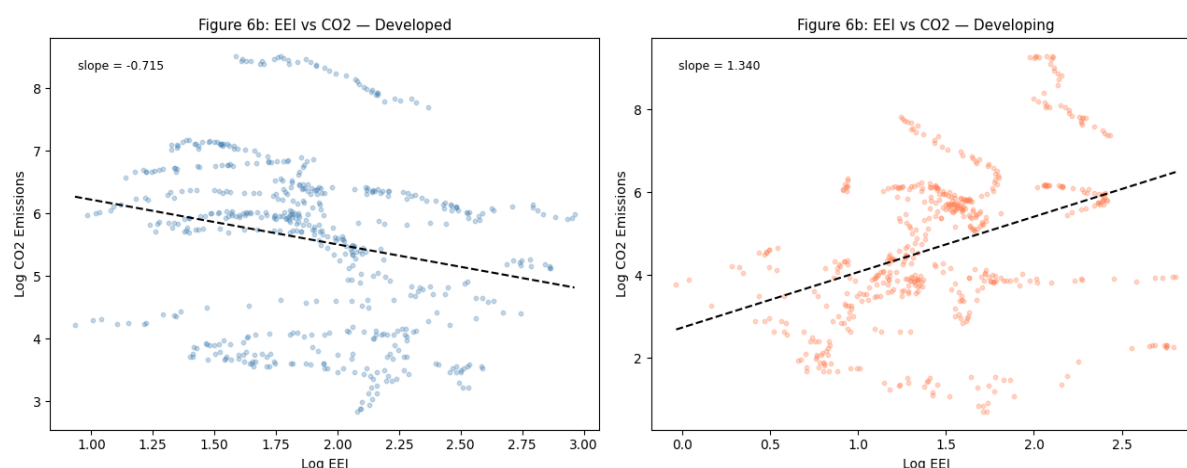


Figure 6b shows scatter plots of log EEI against log CO_2 emissions for developed and developing economies separately. Among developed economies, there is a clear negative relationship (slope = -0.715): as efficiency improves (EEI falls), CO_2 emissions also fall. This is what we would expect, and it shows that efficiency improvements do lead to lower emissions in richer countries. Among developing

economies, the relationship is positive (slope = 1.340). This is because the countries in the developing group with the highest EEI (least efficient) tend to be the most industrialised ones – such as South Africa – which also have high emissions, while the most efficient developing countries (like Kenya) have low emissions simply because they have smaller, less energy-intensive economies.



6.2 Regression Results

Table 4 presents the fixed effects regression results with log CO_2 emissions as the dependent variable and log EEI as the key explanatory variable.

Table 4: Fixed Effects OLS Regression – Dependent Variable: Log CO_2 Emissions

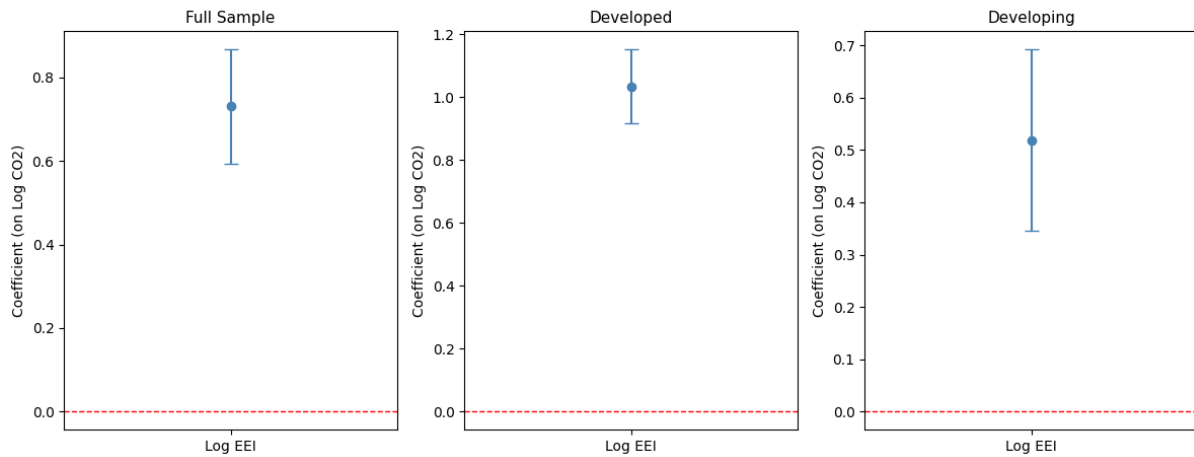
Sample	Coefficient	Std Error	P-value	R^2
Full Sample	0.730	0.070	0.000	0.978
Developed	1.033	0.060	0.000	0.981
Developing	0.519	0.088	0.000	0.981

Note: HC3 heteroscedasticity-robust standard errors. Country and year dummies included. All coefficients significant at $p < 0.001$.

In the full sample, a 1% increase in EEI (meaning efficiency gets worse) is associated with a 0.73% increase in CO_2 emissions. This is a large and statistically significant effect, confirming that energy efficiency and CO_2 emissions are closely linked. The R^2 values of around 0.978–0.981 show that the models fit the data very well.

Importantly, the effect is stronger in developed economies (coefficient = 1.033) than in developing ones (0.519). For developed countries, improving energy efficiency translates into an almost proportionate reduction in CO_2 emissions. For developing countries, the effect is smaller, which is consistent with the idea that economic growth in these countries is generating so much additional energy demand that efficiency gains alone are not enough to bring total emissions down.

Figure 6c: Fixed Effects OLS — EEI on CO₂ (95% CI)



6.3 Discussion and Policy Recommendations (Q7)

The findings suggest that improving energy efficiency is an effective way to reduce CO₂ emissions, but that the effectiveness depends on the stage of economic development. In developed countries, where economic growth is slower and the energy mix is already shifting towards renewables, efficiency gains translate fairly directly into lower emissions. In developing countries, where economies are growing quickly and energy demand is rising rapidly, efficiency improvements alone are not sufficient – they need to be combined with a shift towards cleaner energy sources.

This has important implications for climate policy. For developed economies, energy efficiency standards, carbon pricing, and investment in clean technology should remain central to emissions reduction strategies. For developing economies, the priority should be ensuring that economic growth is powered by clean energy rather than fossil fuels, while efficiency improvements are pursued alongside rather than instead of decarbonisation. International climate agreements such as the Paris Agreement (United Nations, 2015) play a key role in supporting this by encouraging technology transfer and financial support from developed to developing countries.

7. Energy Efficiency in England and Wales

7.1 Regional Distribution of EPC Score (Q1)

This section analyses the distribution of Energy Performance Certificate (EPC) scores across Local Authority Districts (LADs) in England and Wales. EPC scores range from 1 to 100, with higher scores indicating better energy efficiency and lower environmental impact. Scores are grouped into rating bands from A (above 91, most efficient) to G (1–20, least efficient).

7.1.1 Energy Efficiency Scores

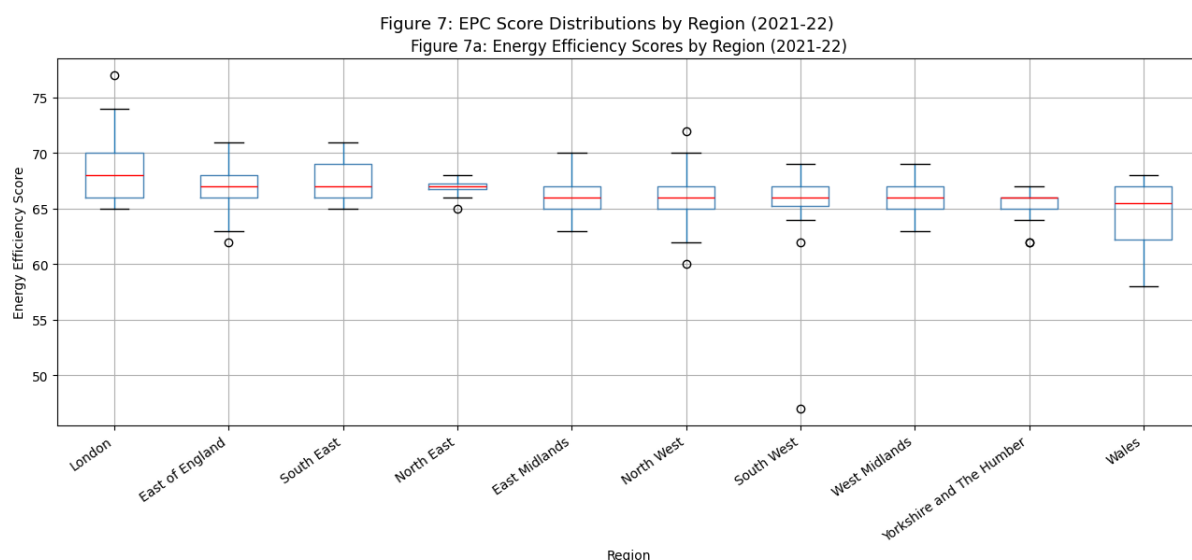
Table 5 shows the distribution of median energy efficiency scores by region for 2021/22. London has the highest median score of 68.0, followed by the South East and East of England (both 67.0). Wales has the lowest median score at 65.5. All regions fall within Band C (69–80), Band D (55–68) or Band E (39–54), and no region contains any Band A or B LADs. This tells us that while energy efficiency across England and Wales is moderate, there is still significant room for improvement, particularly given the UK government’s target of achieving Net Zero emissions by 2050.

Table 5: Median Energy Efficiency Score by Region – 2021/22

Region	Mean	Median	Std Dev	Min	Max
London	68.30	68.0	2.90	65	77
South East	67.39	67.0	1.59	65	71
East of England	66.96	67.0	1.99	62	71
North East	66.92	67.0	0.90	65	68
West Midlands	66.20	66.0	1.47	63	69
East Midlands	65.94	66.0	1.47	63	70
North West	65.76	66.0	2.40	60	72
South West	65.50	66.0	4.13	47	69
Yorkshire & Humber	65.29	66.0	1.64	62	67
Wales	64.45	65.5	3.08	58	68

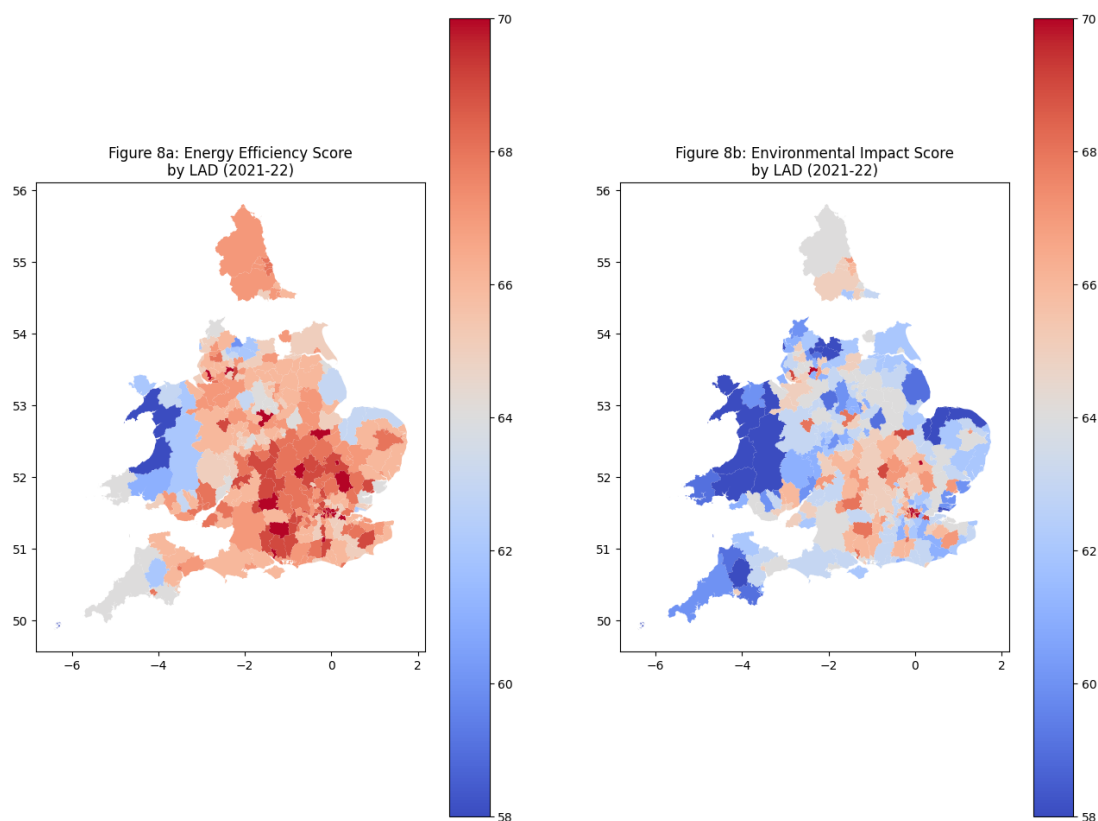
Note: Higher scores indicate greater energy efficiency. Band C = 69–80; Band D = 55–68; Band E = 39–54.

Figure 7a shows box plots of energy efficiency scores by region, ordered from highest to lowest median. London’s distribution is the highest overall but also shows more variation (standard deviation of 2.90), with a high outlier of 77 likely reflecting newer, more energy-efficient urban developments. The South West has the greatest spread of any region (standard deviation 4.13) and the lowest single LAD observation of 47, which likely reflects rural areas with older and harder-to-insulate housing.



Figures 8a and 8b show choropleth maps of energy efficiency and environmental impact scores by LAD. Both maps show a clear spatial pattern: London and the South East tend to score higher (more efficient), while parts of Wales, the South West, and rural northern England score lower. This is likely explained by differences in housing stock age and type, with urban areas containing more newer, better-insulated properties than rural areas.

Figure 8: EPC Scores by LAD (2021-22)



7.1.2 Environmental Impact Scores

Table 6 shows regional summary statistics for environmental impact scores. The pattern is broadly similar to energy efficiency, with London scoring highest (median 66.0) and Wales lowest (median 62.0). One notable difference is that the North East ranks second for environmental impact (65.0) despite ranking fourth for energy efficiency. This suggests that some Northern LADs have relatively low CO₂ emissions per household even though their overall energy efficiency score is lower, which may reflect differences in the type of heating fuel used.

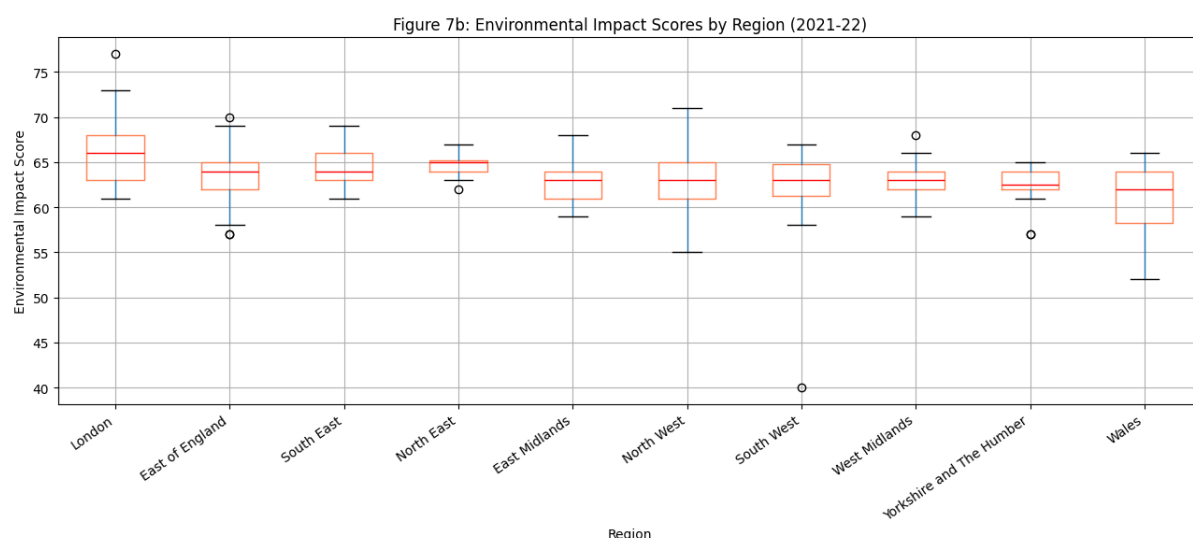
Table 6: Median Environment Impact Scores by Region – 2021/22

Region	Mean	Median	Std Dev	Min	Max
London	65.88	66.0	3.95	61	77
North East	64.67	65.0	1.37	62	67
South East	64.33	64.0	2.14	61	69
East of England	63.64	64.0	2.92	57	70
East Midlands	63.00	63.0	2.00	59	68
North West	62.73	63.0	3.35	55	71
West Midlands	62.97	63.0	1.97	59	68
South West	62.08	63.0	5.07	40	67
Yorkshire & Humber	62.29	62.5	2.55	57	65
Wales	60.86	62.0	4.27	52	66

Note: Higher scores indicate lower CO₂ emissions. Band C = 69–80; Band D = 55–68; Band E = 39–54.

As with energy efficiency, all LADs fall within Bands C, D, or E for environmental impact. No LAD achieves Band A or B. This is a significant finding given the scale of improvement needed to meet the UK's Net Zero target, which will require many buildings to reach at least Band C.

Figure 7b shows the distribution of environmental impact scores across regions in England and Wales for 2021–22. The overall pattern is similar to the energy efficiency scores shown in Figure 7a, with London recording the highest median scores and Wales generally recording lower values. Most regions are concentrated within EPC Bands C and D, indicating moderate environmental performance overall. The South West displays the greatest variation in scores, suggesting larger differences in housing quality and energy usage across local authority districts within the region.



7.2 Changes Over Time (Q2)

7.2.1 Energy Efficiency Score Changes

This section examines whether both energy efficiency and environmental impact scores improved between 2020/21 and 2021/22

Across all 314 LADs, the average energy efficiency score increased by 0.787 points over the year. Of these, 207 LADs (65.9%) improved, 95 (30.2%) stayed the same, and only 12 (3.8%) declined. This shows that progress is being made, though the improvements are small in absolute terms.

Table 7 breaks down the mean change in energy efficiency score by region. Wales and the West Midlands saw the largest average improvements (both around +1.1 points), while Yorkshire and the Humber saw the smallest (+0.21). London, despite having the highest absolute scores, recorded one of the smaller mean improvements (+0.49), suggesting that lower-scoring regions are gradually catching up.

Table 7: Mean Change in Energy Efficiency Score by Region (2020–21 to 2021–22)

Region	Mean Change	Median	Min	Max
West Midlands	+1.10	1.0	0.0	2.0
Wales	+1.09	1.0	0.0	5.0
East of England	+1.02	1.0	0.0	2.0
South East	+0.91	1.0	-1.0	2.0
South West	+0.81	1.0	0.0	2.0
East Midlands	+0.74	1.0	-1.0	2.0
North East	+0.50	0.0	0.0	2.0
London	+0.49	1.0	-5.0	2.0
North West	+0.42	0.0	-1.0	2.0

Yorkshire & Humber	+0.21	0.0	-1.0	1.0
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Note: Positive values indicate a score improvement. Change = 2021/22 score minus 2020/21 score.

Figure 9a plots the average energy efficiency score for each region over both periods. All regions show an upward trend, confirming that improvements are widespread and not concentrated in just a few areas. The lines are broadly parallel, suggesting the pace of improvement is fairly consistent across regions.

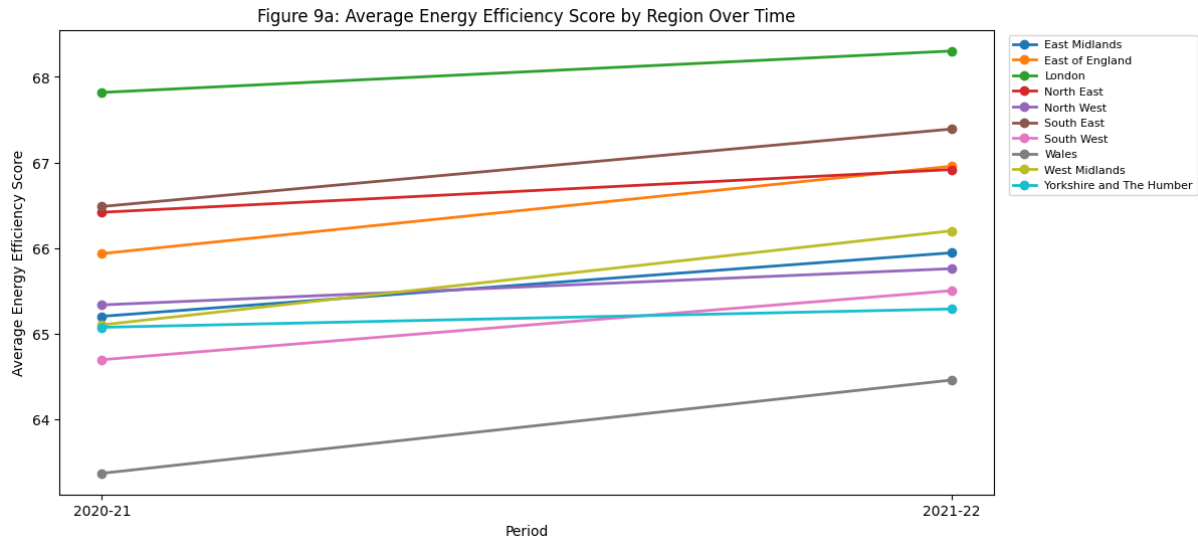
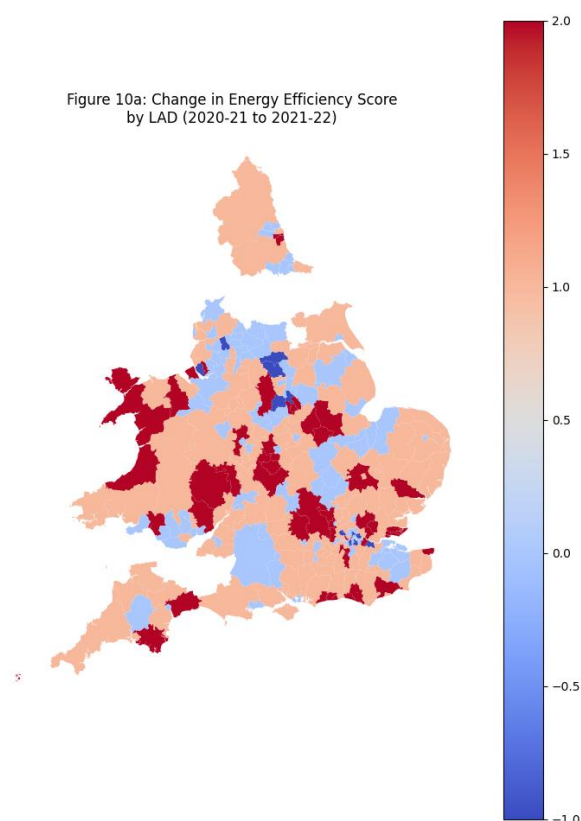


Figure 10a maps the change in energy efficiency score by LAD. The map confirms that improvements are broadly distributed across England and Wales (shown in red/pink), with stronger gains visible in parts of Wales and the West Midlands. The small number of declining LADs (shown in blue) are scattered with no obvious geographic pattern.



7.2.2 Environmental Impact Score Changes

Turning to environmental impact scores, the overall picture is similar to energy efficiency but with some notable differences.

Table 8 shows the mean change in environmental impact score by region. Wales records the largest improvement (+1.27), followed by the West Midlands (+1.17) and East of England (+1.04). These three regions also featured among the top improvers for energy efficiency, suggesting that progress on both dimensions tends to occur in the same areas.

Table 8: Mean Change in Environmental Impact Score by Region (2020–21 to 2021–22)

Region	Mean Change	Median	Min	Max
Wales	+1.27	1.0	-1.0	5.0
West Midlands	+1.17	1.0	0.0	3.0
East of England	+1.04	1.0	-1.0	3.0
South East	+1.03	1.0	-1.0	3.0
South West	+1.00	1.0	0.0	2.0
East Midlands	+0.91	1.0	-1.0	3.0
North East	+0.58	1.0	-1.0	2.0

London	+0.52	1.0	-6.0	3.0
North West	+0.49	1.0	-2.0	2.0
Yorkshire & Humber	-0.14	0.0	-2.0	1.0

Note: Positive values indicate a reduction in CO₂ impact. Change = 2021/22 score minus 2020/21 score

Figure 9b shows the average environmental impact score by region over both periods. All regions show an upward trend except Yorkshire and the Humber, whose line is almost flat, consistent with its recording the only negative mean change at the regional level. London starts and ends the highest, while Wales starts the lowest but shows one of the steepest upward lines, consistent with its recording the largest regional mean improvement.

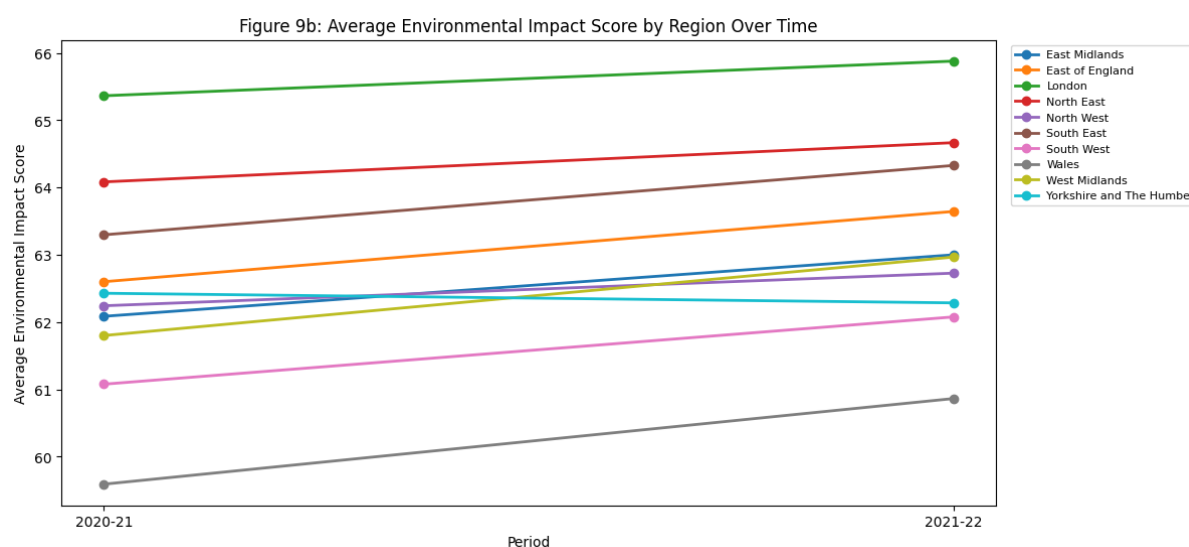


Figure 10b maps the change in environmental impact score by LAD. Compared to the energy efficiency change map, there are visibly more blue patches – indicating declines – particularly around London and parts of Yorkshire. This is consistent with the regional finding that Yorkshire and the Humber recorded a negative mean environmental impact change despite a small positive energy efficiency improvement. Wales and the West Midlands again show strong gains (deep red), while much of the South East and East of England shows moderate progress.

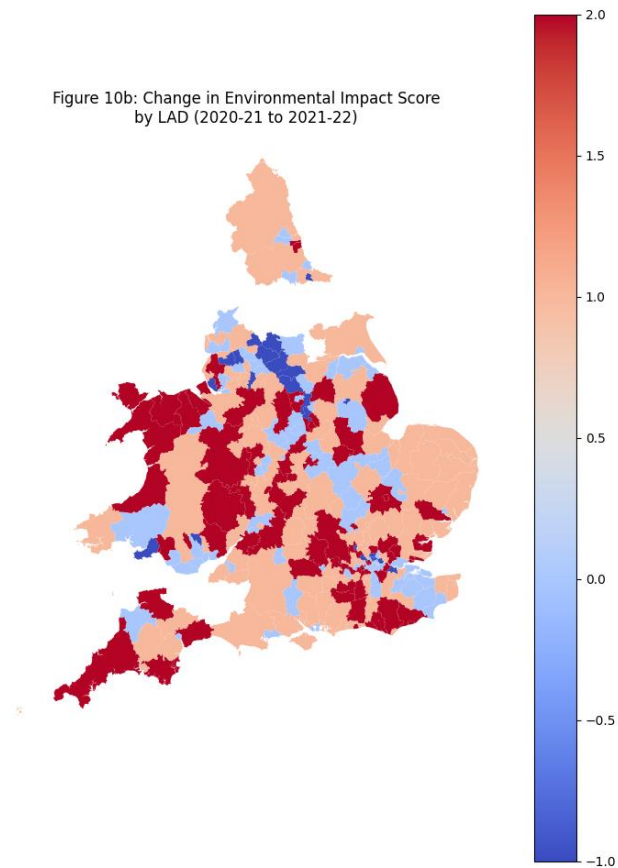
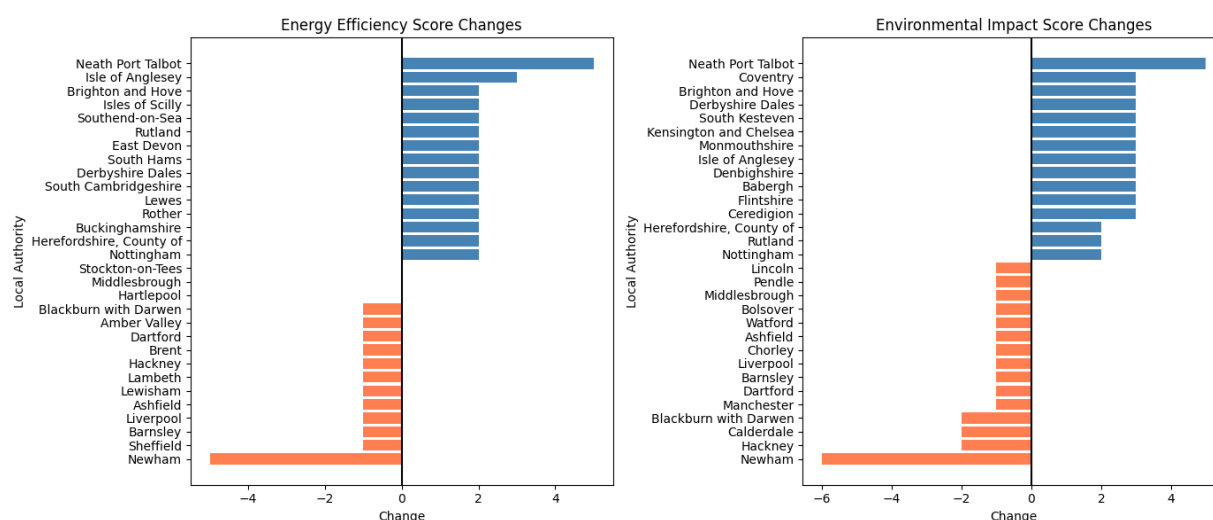


Figure 11 presents the 15 LADs with the largest improvements and the 15 with the largest declines in both energy efficiency and environmental impact scores side by side. Neath Port Talbot in Wales records the largest improvement in both measures, which is consistent with Wales showing one of the biggest average regional improvements overall. In contrast, Newham in London records the largest decline, with scores falling by around 5 points for energy efficiency and 6 points for environmental impact. Since Newham appears among the largest declines in both measures, this suggests a more consistent deterioration rather than a one-off issue affecting only a single indicator. Sheffield also records noticeable declines, although the majority of LADs continue to show either modest improvements or little change over the period.



Overall, both energy efficiency and environmental impact scores improved across most LADs between 2020/21 and 2021/22. However, the pace of improvement is modest, regional gaps persist, and Yorkshire and the Humber is the only region where environmental impact did not improve on average. More targeted policy intervention in underperforming areas will be needed to accelerate progress towards the UK's Net Zero 2050 target.

8. Conclusion

This report has examined global trends in energy efficiency between 1980 and 2019, the economic factors that drive those trends, the relationship between energy efficiency and CO_2 emissions, and the performance of LADs in England and Wales using EPC scores.

The main findings are as follows. Global energy efficiency has improved significantly over the study period, with the average EEI falling by over 43%. Both developed and developing economies have contributed to this improvement, and the gap between the two groups has narrowed over time. The United Kingdom performs well by global standards, ranking 6th out of 28 countries in 2019.

The regression analysis shows that renewable energy share and GDP per capita are the most important drivers of energy efficiency in developed economies, while trade openness is the key factor for developing economies. This suggests that policymakers in developed countries should continue to invest in clean energy and enforce energy efficiency standards, while developing countries would benefit most from policies that improve access to cleaner imported technologies.

On the relationship between energy efficiency and CO_2 emissions, the results confirm that improving efficiency does reduce emissions, but the effect is stronger in developed economies than in developing ones. In developing countries, economic growth and rising energy demand mean that efficiency gains alone are not enough to

reduce total emissions – they need to be paired with a move towards cleaner energy sources.

In England and Wales, all LADs fall within EPC Bands C to E, with London performing best and Wales scoring lowest on average. The majority of LADs improved their scores between 2020/21 and 2021/22, which is a positive sign, but the pace of improvement is modest and will need to accelerate significantly to meet the UK's Net Zero 2050 target.

There are some limitations to this analysis. The global sample of 28 countries, while geographically diverse, does not represent all economies. The EPC analysis covers only two years, which limits the ability to identify longer-term trends. Future work could extend the time period and incorporate additional variables such as energy prices to get a more complete picture.

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